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08 November 2025 07:38

6. Presentation Layer (Layer 6)

- **Function**: Prepares data for the application layer, handling translation, encryption, and compression.
- **Responsibilities**:
  - Translates data between application and network formats (e.g., encoding files).
  - Compresses data to reduce size for faster transmission.
  - Encrypts data to secure it during transmission.
- **Examples**: Converting images to standard formats (JPEG, PNG) for transmission.
- **Protocols/Technologies**: SSL/TLS (for encryption), JPEG, GIF, ASCII, EBCDIC (for encoding).

7. Application Layer (Layer 7)

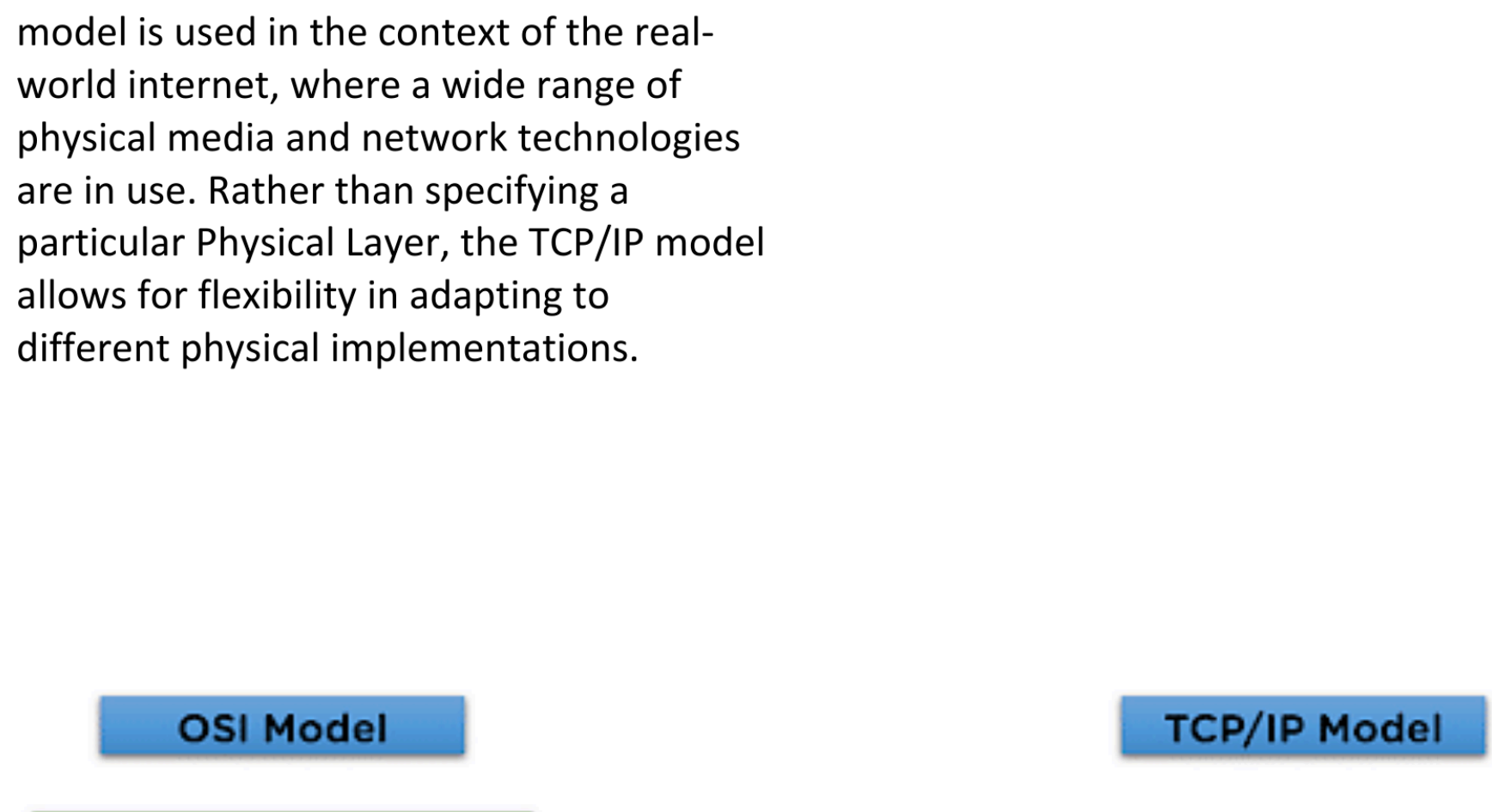
- **Function**: Interfaces directly with end-user applications and facilitates network services.
- **Responsibilities**:
  - Provides services to applications (like email, file transfer, web browsing).
  - Facilitates network access, data retrieval, and error handling for end-user applications.
  - Acts as the layer where users interact with the network through applications.
- **Examples**: A web browser requesting a web page via HTTP.
- **Protocols/Technologies**: HTTP/HTTPS (web browsing), FTP (file transfer), SMTP (email), DNS (domain name resolution).

TCP/IP Model Overview

The **TCP/IP Model** (Transmission Control Protocol/Internet Protocol) is a four-layer networking model that describes how data is transmitted over networks, including the internet. Unlike the OSI model, which is conceptual, the TCP/IP model is practical and widely implemented, serving as the foundation of internet protocols.

What Does TCP/IP Do?

The main work of TCP/IP is to transfer the data of a computer from one device to another. The main condition of this process is to make data reliable and accurate so that the receiver will receive the same information which is sent by the sender. To ensure that, each message reaches its final destination accurately, the TCP/IP model divides its data into packets and combines them at the other end, which helps in maintaining the accuracy of the data while transferring from one end to another end. The TCP/IP model is used in the context of the real-world internet, where a wide range of physical media and network technologies are in use. Rather than specifying a particular Physical Layer, the TCP/IP model allows for flexibility in adapting to different physical implementations.



Layers of the TCP/IP Model:

Application Layer
Transport Layer
Internet Layer
Network Access Layer

1. Network Access Layer (Link Layer)

- **Function**: Manages the physical transmission of data across network media.
- **Responsibilities**:
  - Handling communication between devices on the same network (e.g., within a LAN).
  - Packaging data into frames for transmission.
  - Managing MAC (Media Access Control) addressing to identify devices on a local network.
- **Protocols**: Ethernet, ARP (Address Resolution Protocol), PPP (Point-to-Point Protocol).
- **Example**: When a device sends data within a LAN, the Link Layer handles the actual connection to the physical media and local delivery.

2. Internet Layer (Network Layer)

- **Function**: Responsible for logical addressing and routing of packets across networks.
- **Responsibilities**:
  - Determining the best path for data to reach its destination (known as routing).
  - Using IP addresses to uniquely identify devices on a network.
  - Fragmenting and reassembling packets as they pass through different networks.
- **Key Protocols**:
  - **IP (Internet Protocol)**: Assigns IP addresses and routes packets between devices.
  - **ICMP (Internet Control Message Protocol)**: Sends error messages and operational information (e.g., "ping" uses ICMP).
  - **IGMP (Internet Group Management Protocol)**: Manages multicast group memberships.
- **Example**: When data needs to travel across different networks, the Internet Layer finds the correct path for each packet and forwards it accordingly.

3. Transport Layer

- **Function**: Provides end-to-end communication and ensures data integrity.
- **Responsibilities**:
  - Managing communication between two devices over the network, ensuring reliable data transfer.
  - Segmenting large data into smaller chunks, known as segments, for efficient transfer.
  - Reassembling segments at the destination and ensuring all data is received correctly and in order.
- **Key Protocols**:
  - **TCP (Transmission Control Protocol)**:
    - Connection-oriented protocol ensuring reliable data transfer.
    - Includes mechanisms for error checking, flow control, and retransmission of lost packets.
  - **UDP (User Datagram Protocol)**:
    - Connectionless protocol providing faster, less reliable transmission.
    - Often used in applications where speed is more critical than reliability (e.g., live video streaming).
- **Example**: When you download a file, TCP at the Transport Layer ensures that all parts of the file arrive correctly. If any part is missing, TCP requests retransmission.

4. Application Layer

- **Function**: Facilitates communication between network applications and provides a user interface.
- **Responsibilities**:
  - Enabling applications to access network resources.
  - Handling high-level protocols that allow software to send and receive data over the network.
  - Serving as the interface between network services and applications.
- **Key Protocols**:
  - **HTTP/HTTPS (Hypertext Transfer Protocol)**: Used for browsing web pages.
  - **FTP (File Transfer Protocol)**: For transferring files.
  - **SMTP (Simple Mail Transfer Protocol) and IMAP/POP3**: For email services.
  - **DNS (Domain Name System)**: Resolves domain names to IP addresses.
- **Example**: When you enter a URL in a browser, the Application Layer uses HTTP to request and receive web page data.

How the Layers Work Together

In the TCP/IP model:

1. The **Application Layer** prepares data for sending, using protocols like HTTP or FTP.
2. The **Transport Layer** breaks down this data into segments (TCP) or datagrams (UDP) and ensures reliable or fast transmission.
3. The **Internet Layer** attaches an IP address to each segment, determining the route for the packet to reach the correct destination.
4. The **Link Layer** sends the data to the next device on the network, handling the physical connection and local addressing.

This layered approach allows TCP/IP networks to be robust, flexible, and scalable, ensuring devices from different manufacturers and systems can communicate effectively.

Example : **Sending a text message "Hello" from one phone to another**.

1. **Application Layer**: Prepares "Hello!" to be sent over the network. When you type "Hello!" in your messaging app and press send, the Application Layer prepares this message for delivery by converting it to a format suitable for network transmission.

2. **Transport Layer**: Breaks "Hello!" into numbered segments, ensuring they'll be delivered fully and in order.

3. **Internet Layer**: It decides the best path for each data packet to travel across different networks to reach the other phone. Each segment of your message is given an IP address (like a home address) so it knows where to go. This layer uses routers to move packets across different networks until they reach the recipient's IP address.

4. **Link Layer**: Transmits each packet physically from device to device across networks until they reach the destination.

When the other phone receives all packets, the process reverses to reassemble the original message. The user on the other phone then sees "Hello!" just as it was typed!

TCP vs IP

TCP (Transmission Control Protocol) and **IP** (Internet Protocol) are **not the same**, but they work together closely as part of the **TCP/IP model** to enable reliable communication over the internet.
Here's how they differ and complement each other:

1. IP (Internet Protocol)

- **Purpose**: IP is responsible for addressing and routing packets of data between devices across networks.
- **Function**: Think of IP as a **postal address system**. It attaches an IP address to each data packet, which helps guide it through networks to reach the correct destination.
- **Reliability**: IP is **unreliable by itself**. It doesn't guarantee that data will arrive at the destination, arrive in order, or even arrive at all. It only ensures data is sent to the correct address.

2. TCP (Transmission Control Protocol)

- **Purpose**: TCP is responsible for ensuring **reliable, ordered, and error-checked delivery** of data between devices.
- **Function**: Think of TCP as a **reliable delivery service** that checks that each part of a message has arrived, resends any lost parts, and puts everything back in the correct order.
- **Reliability**: TCP adds reliability to the communication by confirming that each data segment has arrived and, if not, resending it. This ensures that the data reaches the destination accurately and completely.

How TCP and IP Work Together

- **IP** routes data to the correct destination using IP addresses.
- **TCP** makes sure the data reaches fully and correctly by checking for errors and resending missing parts.

Together, **TCP/IP** provides a complete system for sending data over the internet—**IP** handles **addressing and routing**, while **TCP** ensures **reliability and completeness**.

3. IP Addressing

IP Addressing is the system used to uniquely identify devices on a network, allowing them to communicate with each other. An IP (Internet Protocol) address is like a postal address for networked devices, helping data packets find their way from the sender to the receiver.

What is an IP Address?

An **IP address** is a unique identifier assigned to each device connected to a network. It allows devices like computers, phones, and printers to communicate with each other. Think of it as a device's "home address" in a network, guiding data to the right destination.

Types of IP Addresses

There are two main versions of IP addresses:

1. **IPv4 (Internet Protocol Version 4)**
2. **IPv6 (Internet Protocol Version 6)**

I. IPv4 (Internet Protocol Version 4)

IPv4 is the older and most commonly used IP address format. In IPv4, the IP address are displayed in dotted-decimal notations, and appear as a four numbers separated by a dot. Each number of an IPv4 address is made from eight individual bits known as Octet. Each octet can have a value from 0 to 255, allowing for about 4.3 billion unique addresses.
Example of IPv4 Address: 192.168.0.1

Structure of an IPv4 Address

An IPv4 address is divided into **four octets** (or bytes) separated by dots, each ranging from 0 to 255 (e.g., 192.168.1.1).

- **Network Part**: Identifies the specific network on which the device is located.
- **Host Part**: Identifies the specific device within that network.

The ratio of network to host portions depends on the **IP class** or **subnet mask**.

IP Address Classes in IPv4

IP address classes are a way of categorizing IP addresses to help manage networks of different sizes. In IPv4, the addresses are divided into five classes: **Class A, B, C, D, and E**, but we mainly use **Class A, B, and C** for assigning IPs to devices. Each class has a range of IP addresses and is used for different network sizes.
Here's an easy breakdown of each class:

Class A

- **Range**: 1.0.0.0 to 126.0.0.0
- **Structure**: The first **8 bits** (1st octet) represent the **network** part, and the remaining **24 bits** (3 octets) are for the **host** part.
- **Number of Networks**: 128 networks (actually fewer due to reserved IPs).
- **Hosts per Network**: ~16 million devices.
- **Usage**: Designed for **very large networks**, such as big corporations or ISPs.

Example: 10.0.0.1 could be used within a Class A network.

Class B

- **Range**: 128.0.0.0 to 191.255.0.0
- **Structure**: The first **16 bits** (2 octets) represent the **network** part, and the remaining **16 bits** are for the **host** part.
- **Number of Networks**: 16,384 networks.
- **Hosts per Network**: ~65,000 devices.
- **Usage**: Suitable for **medium-sized networks**, like universities or smaller companies.

Example: 172.16.0.1 could be an address in a Class B network.

Class C

- **Range**: 192.0.0.0 to 223.255.255.0
- **Structure**: The first **24 bits** (3 octets) represent the **network** part, and the remaining **8 bits** are for the **host** part.